



# 5G Edge AI Lab

UE → RAN → Core → Edge Kubernetes → GPU/VLM, end to end  
on one machine

`Open5GS · UERANSIM · minikube · llama.cpp + Qwen2-VL`

[github.com/tarekabouzeid/edge-compute-open5gs-lab](https://github.com/tarekabouzeid/edge-compute-open5gs-lab)

## What the lab answers

### Where should AI inference sit in a 5G network — and how big must the edge GPU be?

Everything from signalling to GPU inference runs on real, production-grade open-source software. Only the radio is simulated.

1

#### Real 5G core

Open5GS: 11 network functions, NAS/NGAP signalling, PDU sessions, GTP-U tunnelling

2

#### Real edge compute

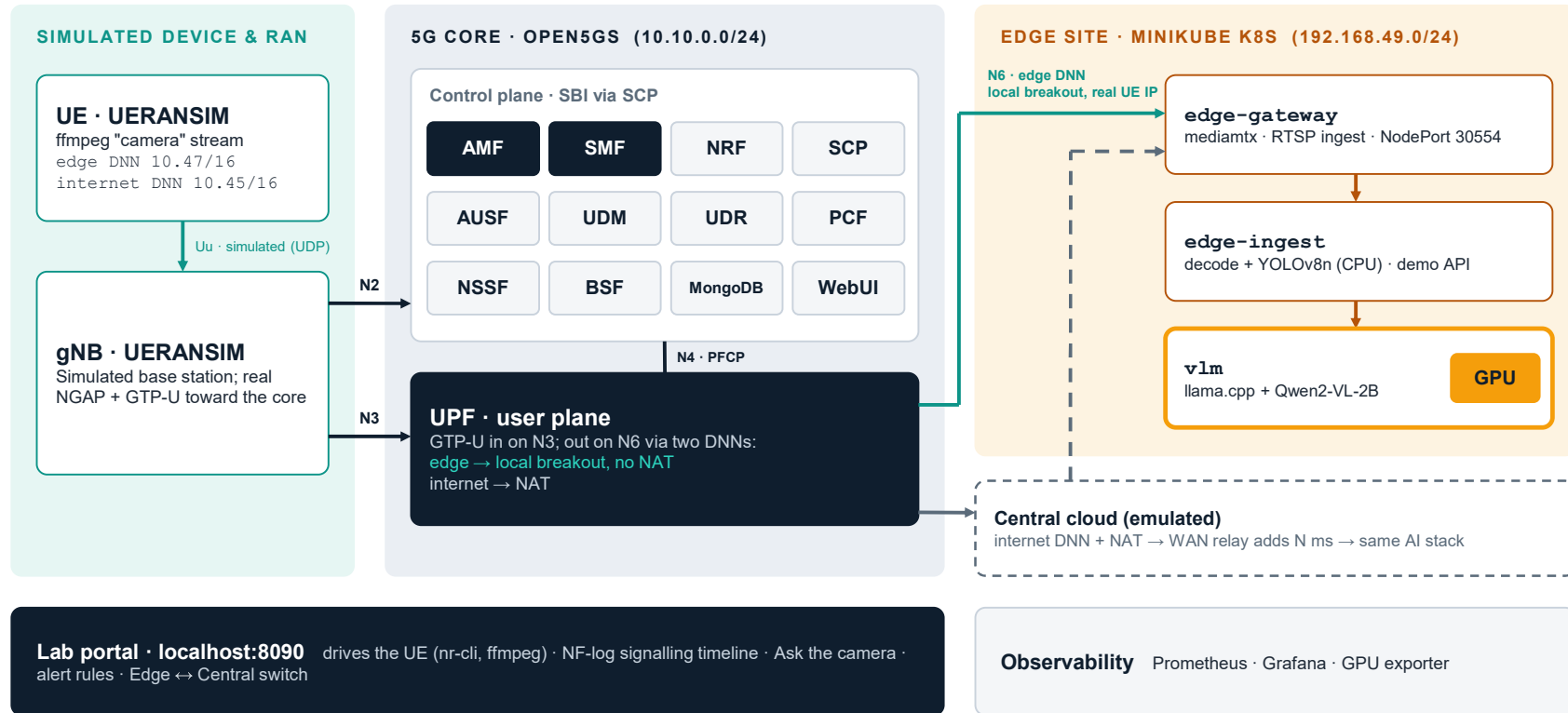
minikube with the host GPU; YOLOv8n object detection and a Qwen2-VL-2B vision-language model

3

#### Simulated radio only

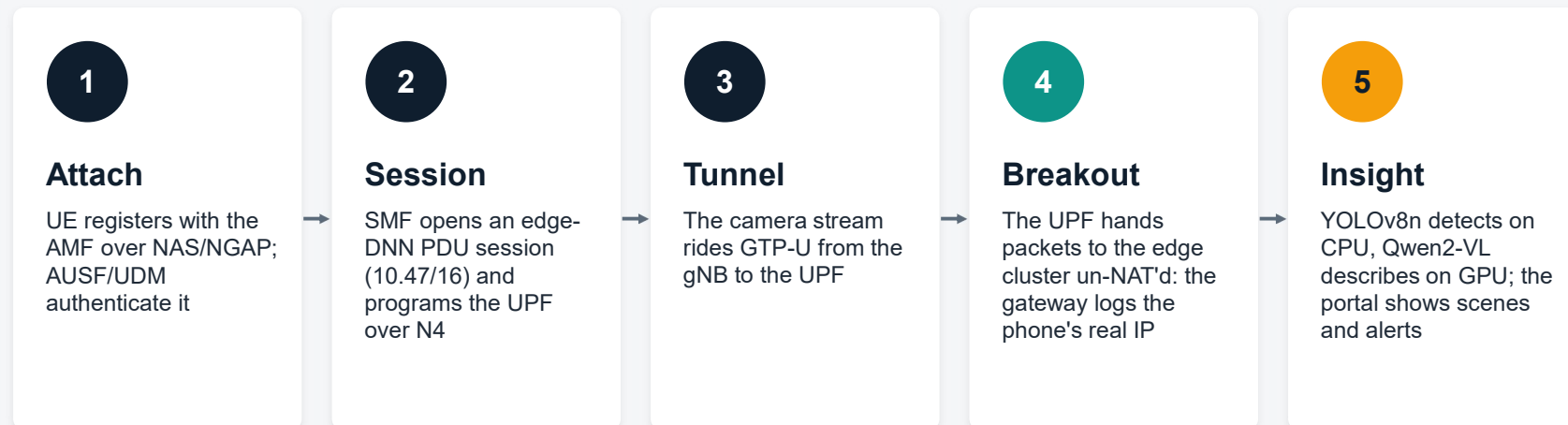
UERANSIM carries the RF/PHY layer over UDP: no SDR hardware, no radio licence

# High-level architecture



One Docker host · WSL2 Ubuntu 24.04 · NVIDIA RTX 5070 Ti · every component version-pinned

## One video frame, end to end



**Edge vs central, measured.** The portal's switch moves the stream onto the internet DNN and a WAN-delay relay, then measures round-trip and time-to-insight from inside the UE on both paths.

# What is real, what is simulated

| Layer                                       | Status    | How                                                        |
|---------------------------------------------|-----------|------------------------------------------------------------|
| 5G core signalling: NAS, NGAP, PDU sessions | Real      | Open5GS, same 3GPP interfaces as a commercial core         |
| IP data path                                | Real      | Real packets, TUN interfaces, GTP-U encapsulation          |
| Edge Kubernetes + GPU scheduling            | Real      | minikube with host GPU (K3s + GPU Operator as alternative) |
| Video ingest + inference                    | Real      | YOLOv8n on CPU; llama.cpp + Qwen2-VL-2B on GPU             |
| Phone camera                                | Simulated | Real video file streamed by ffmpeg inside the UE container |
| Radio / PHY                                 | Simulated | UERANSIM over UDP: the one deliberate gap                  |
| Central cloud                               | Emulated  | Real NAT path + relay adding configurable WAN delay        |

## Why this scope

The question is compute placement and GPU sizing, not radio engineering.

Real RF needs SDR hardware and a licence or shielded room, and would not change the latency, throughput or saturation answers.

## Stack and status

### RAN

UERANSIM gNB + UE

### CORE

Open5GS · 11 NFs + MongoDB

### EDGE

minikube · NVIDIA device plugin

### AI

YOLOv8n · Qwen2-VL-2B

### OPS

Prometheus · Grafana · lab portal

### CI

Lint · kubeconform · core smoke

**Running end to end** on WSL2 + RTX 5070 Ti, with verified evidence for every layer.

```
$ ./lab.sh all up
```